

Algorithmic imaginaries: definition and critical questions

Dr Andreas Schellewald

Dr Ellenrose Firth

Dr Hui Lin

Sense-making about algorithms / AI systems is now a crowded conceptual field

- Many concepts compete: **mental models, folk theories, algorithmic imaginaries**
- Adjacent work on awareness, literacy, aversion, conspiritoriality, knowledge
- Each foregrounds different empirical and analytical commitments

Algorithms to AI Shift

Risk of re-inventing concepts rather than refining what a decade plus of scholarship already yields in utility.

The algorithmic imaginary offers unique utility value as a sensitising concept

Definitive (Blumer 1954)

Fixed attributes; classify instances.

Mental models and **folk theories** sit here, providing clear cut definitions and typologies of different theories and models.

Sensitising

Directions to look, not what to see.

The **algorithmic imaginary** sits here, foregrounding dimensions of affect, emotion, process in research on human-algorithm interaction.

Bucher's imaginary concept theorises sense-making as an affective and idiosyncratic process

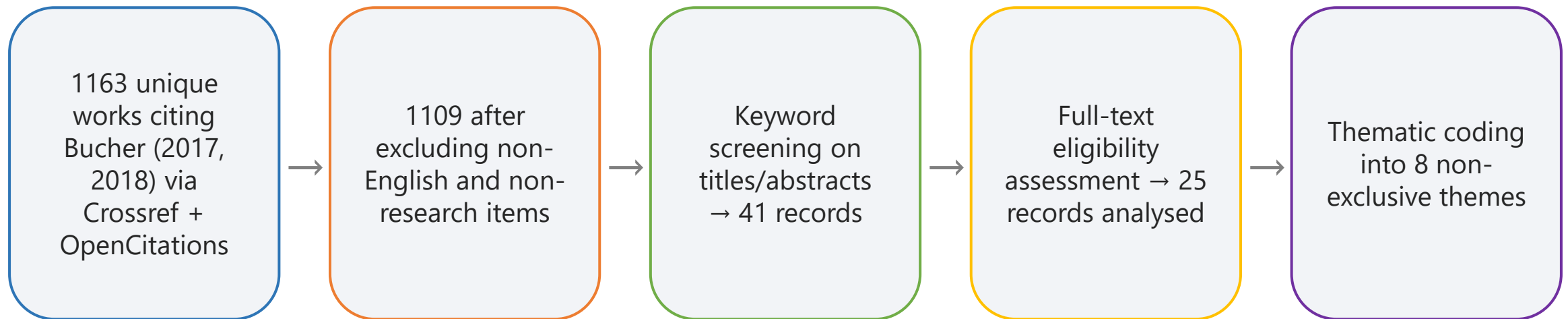
- A figurative space where **algorithms and users meet** (Bucher 2017)
- Productive, generative force from **situated, affective encounters**
- Not a belief held, but a recursive shaper of interactive opportunity, practice, and live experience



Conceptualising Imaginaries

Distinct from data, platform, sociotechnical, or social imaginaries — which operate and theorise imagination at institutional and collective scales, rather than individual idiosyncrasies.

Synthesising a decade of explicit engagement with the algorithmic imaginaries concept



Eight themes, grouped into three critical threads



Power

Political and cultural engagements – who shapes imaginaries, how and whose values are encoded in them.



Collectivity

Collective, affective, contextual — imaginaries as shared, situated, socially orchestrated manifestations.



Agency

Agentic and technical — imaginaries as action-spaces; interface design as imaginative surface for interactivity and research.

Power: the literature has outlined multiple angles to see how imaginaries intersect with questions of power

- **Designers' imaginations** shape user experience: e.g. Alexa's affective branding (Kopitz 2021)
- Imaginaries can **constrain** — creators vulnerable to algorithms imagined as objective arbiters (Hao 2024; Wang & Cao 2024)
- They also **catalyse critique**: mutual aid, counter-publics, coordinated resistance (Maris et al. 2024; Jones 2023)



Critical question

How and by whom are imaginaries shaped?
How do they reproduce power dynamics?

Power: the role of values in mediating which encounters become forceful irritations

- Algorithms encode **institutionalised knowledge logics** (Gillespie 2014)
- Mismatch with community values makes algorithms **visible**
- Cases: BookTok literacy (Low et al. 2023), Douyin entrepreneurialism (Wang and Cao 2024), TikTok mutual aid (Maris et al. 2024).

Critical question

What is the role of algorithmic imaginaries in mobilising critical engagement with that power?

Collectivity: algorithmic imaginaries can be the result of collective experience and sense-making

- Memes, fandoms, faith communities imagine algorithms **together** (e.g., Reinis & Laughlin 2025)
- People generalise personal encounters as a shared '**algorithmic other**' (Gandini et al. 2023)
- Extends Bucher's individual-situated framing toward group sense-making

Critical question

What are mechanisms of collective sense-making, shared emotional experience, and affective irritation that produce shared imaginaries?

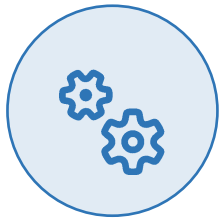


An update definition with three shifts

Algorithmic imaginaries are ... a societal playing field where humans and machines are socially and culturally configured as interaction partners. Shaped by historically specific ways of designing, thinking about, and emotionally relating to technology.

- **From meeting-space to playing field**: stronger emphasis on role of power and values
- **From algorithms to machines**: opening up to Communicative AI, LLMs, similar systems
- **Historically specific**: e.g., ELIZA ≠ ChatGPT – imaginaries are temporally situated

Three critical directions for future research on and around algorithmic / AI imaginaries



Process

How does the imagination process unfold – and what influences and dynamics shape it?



Collectivity

When do people enter the imaginary as a shared space, and how is collective force directed?



Culture

How do norms, values, beliefs position humans and machines, alongside affect?

✓ Conclusions

- The algorithmic imaginary's conceptual value is **sensitising**, not definitive
- Updated definition bridges **individual affect and collective culture**
- Aim to preserve scholarship rather than re-invent for **AI imaginaries** future
- Centre **values, power, and historical specificity** through the concept in future research on human–machine relations

Thank you

Dr Andreas Schellewald (aschellewald@acm.org) | Dr Ellenrose Firth | Dr Hui Lin